Alex Tyh Wang

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Summary

Computer science undergraduate seeking a software engineering internship/co-op for Spring 2027, with experience building AI developer platforms, maintaining CI/CD pipelines, and improving the cost and performance of production AI systems.

Education

Rochester Institute of Technology — Rochester, NY
Bachelor of Science in Computer Science — Expected May 2028

Skills

Languages: Python, TypeScript, JavaScript, SQL, C, C++, C#, Java
Platform and Backend: Jenkins, CI/CD, Azure DevOps, asyncio, REST APIs, ASP.NET, PostgreSQL, SQLite, Cloudflare R2
AI Systems: AI agents, MCP, tool calling, context engineering, LLM evaluation, subagent architectures, token optimization
Web: Next.js, Angular, React, Node.js, server-sent events (SSE), IndexedDB

Experience

Software Engineer Intern — Jan 2026 – Aug 2026
Paychex — *Rochester, NY*

- Led development of an internal AI developer platform on the **AI Integrations** team that integrated **AI agents and MCP tooling** into enterprise developer workflows; presented the platform to executives.
- Built a custom **LLM evaluation harness** for **developer and Sales workflows** to assess model reliability and usefulness.
- Reduced token usage by **55%** with a **subagent architecture** while improving internal agent workflow performance.
- Maintained and updated **Jenkins CI/CD pipelines** supporting automated build and deployment workflows.

Student Researcher — Jan 2026 – May 2026
Rochester Institute of Technology — *Rochester, NY*

- Conducted research on **large language models and memory**.
- Focused on how memory and context shape long-term model behavior.

Software Engineer Intern — Feb 2025 – Aug 2025
Real Wave Capital — *Remote*

- Integrated OpenAI API and OpenRouter into production agent flows with **.NET backend** contributions, improving endpoint response times by **25%** and scaling to **6,000+ weekly requests**.
- Architected Angular and TypeScript UIs for AI agents serving **100+ daily users**; rewrote **MCP tool-calling and context strategy**, reducing token spend by roughly **50%**.

Projects

iMessages Chatbot Server — Nov 2025 – 2026
Python, asyncio, SQLite, AppleScript — *github.com/alextyhwang/iMessages-Chatbot-Server*

- Built a **Python asyncio API gateway** for macOS Messages supporting up to **20 concurrent stateful AI conversations**, with semaphore-controlled API calls and locks for sequential message delivery.
- Implemented **SQLite conversation memory** and **0.3-second debounce batching** to preserve context and prevent premature AI requests when users send rapid messages.

Chatio Agentic Behavior Benchmark and Chat App — Jun 2025 – Present
Next.js, Python, TypeScript, Cloudflare R2 — *chatio.dev*

- Created a **streaming AI chat interface** with SSE, multi-step tool-calling workflows, and dynamic parameter optimization.
- Engineered persistence with **IndexedDB and Cloudflare R2**, reducing image storage by **95%** through compression.
- Built an **LLM evaluation pipeline** combining human review, ensemble model judging, and deterministic Python checks.

Leadership and Activities

Founder and President — Jan 2022 – Present
Bay Area Stem Academy — *San Jose, CA | bayareastemacademy.org*

- Founded a 501(c)(3) nonprofit that taught **660+ students** through **15+ free camps**, partnered with **5+ schools**, and delivered **$35,000+** in free STEM education.
- Led curriculum and operations, donated **$5,000+** to partner organizations and teams, and organized a **SanDisk** volunteer day to build STEM kits for future programs.

Event Coordinator and Executive Board Member — Sep 2024 – Present
RIT Asian Culture Society — *Rochester, NY*

- Co-lead a **200+ member** community and help organize Night Market, an annual event with **600 attendees**.